

RUP Phase Artefacts Master List

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 |
Research/RUP_Phase_Artefacts.md

Overview

Resilio followed the Rational Unified Process (RUP) methodology across four phases. This document lists every artefact produced in each phase, its purpose, and where it appears in the project record. Cross-references to the weekly artefact logs in `Project Artefact/` are provided for each item.

PHASE 1: INCEPTION (Weeks 1–4, November 2024)

Objective: Establish the business case, define the scope, identify stakeholders, assess feasibility, and set up the project environment.

Artefacts Produced

Artefact	Description	Evidence Location
Project Proposal	Formal project proposal defining the problem, proposed solution, and project scope	<code>`Project Artefact/01 Feasibility Study/`</code>
Vision Document	High-level statement of what Resilio is, who it is for, and what success looks like	Chapter 1 of FYP report
Feasibility Study	Technical, economic, operational, and schedule feasibility assessment	<code>`Project Artefact/01 Feasibility Study/`</code>
Stakeholder Register	Identification of all stakeholders: students, therapists, admin users, external clients	Chapter 3, Section 3.2
Pre-Survey Instrument	10-question Google Forms survey distributed to 166 respondents	Appendix G; <code>`Surveys.md`</code>

Artefact	Description	Evidence Location
Pre-Survey Results	Aggregated findings from 166 responses validating the problem domain	Chapter 1, Section 1.3; Appendix G
Risk Register	Identification of technical, schedule, and domain risks and mitigation strategies	Chapter 3, Section 3.3
Initial Gantt Chart	Planned project schedule across all four RUP phases	Appendix J.1
Work Breakdown Structure	Top-level decomposition of all project tasks	Appendix J.2
Client Identification	Identified Ms. Sushana Karki and Dr. Dibyandra Singh as external clients	Appendix F
Technology Selection	Initial selection of Flutter, Node.js, Supabase, SuperTokens based on feasibility	Chapter 3, Section 3.4
Mind Map	Full project scope visualisation	Appendix J.3

Key Decisions Made in Inception

- **Methodology:** RUP selected over Agile, Waterfall, and Prototype (see Appendix E for comparison matrix)
- **Authentication:** SuperTokens OTP selected over Firebase Auth for passwordless local-friendly authentication
- **Payment:** eSewa selected as the only payment gateway due to Nepal market accessibility
- **Platform:** Android-first deployment; iOS deferred to post-submission due to App Store compliance timeline

PHASE 2: ELABORATION (Weeks 5–8, December 2024)

Objective: Refine requirements through stakeholder engagement, produce the system architecture, create design artefacts, identify and mitigate high-risk components through prototyping, and deliver the Interim Report.

Artefacts Produced

Artefact	Description	Evidence Location
Interim Report	Full academic report covering background, requirements, initial design, and methodology	`FYP Report Writing/Interim Report/`
Software Requirements Specification (SRS)	Complete functional and non-functional requirements in SRS format	Appendix M
Use Case Diagrams (x3)	Customer, Therapist, and Admin use case diagrams	Appendix J.5; `UML Diagrams (Mermaid Code).md`
Entity-Relationship Diagram	Full database schema with all entities and relationships	Appendix J.8
System Architecture Diagram	Three-tier architecture: Flutter → Node.js → Supabase	Appendix J.4
Activity Diagrams (x3)	Appointment booking, WebRTC consultation, content publication flows	Appendix J.6
Sequence Diagrams (x3)	OTP login, appointment booking, WebRTC session establishment	Appendix J.7
Data Flow Diagrams (Level 0 + Level 1)	Context diagram and detailed customer/therapist/admin flows	Appendix J.9
Wireframes (x7 screens)	Hand/tool-drawn wireframes for all major screen categories	Appendix J.10
Figma UI Mockups	High-fidelity UI designs for all screens	Appendix J.11
SuperTokens Integration Prototype	Proof-of-concept validating OTP authentication flow	Week 6 artefact log
WebRTC Integration Prototype	Proof-of-concept validating peer-to-peer video connection via Socket.IO signalling	Week 7 artefact log
eSewa Integration Prototype	Proof-of-concept validating payment initiation and callback verification	Week 8 artefact log
Client Requirement Sessions	Structured interviews with Ms. Sushana Karki and Dr. Dibyandra Singh	Appendix F; `05_Client_and_Stakeholder_Documents.md`
Database Schema (PostgreSQL)	Full table definitions, constraints, RLS policies	Appendix M; `08_Database_Schema.md`

Key Decisions Made in Elaboration

- **Clean Architecture:** Feature-first Clean Architecture adopted for Flutter codebase (Martin, 2017) to enable scalable parallel development of Customer, Therapist, and Admin modules
- **State Management:** Flutter BLoC selected for predictable state management across all modules
- **Offline Strategy:** SQLite (sqflite) selected for local persistence; Protocol Buffers for serialisation; LRU cache for content
- **Signalling:** Socket.IO selected for WebRTC signalling due to mature Flutter SDK and reliability
- **Content delivery:** Cloudinary selected for media hosting due to free tier and CDN delivery

PHASE 3: CONSTRUCTION (Weeks 9–19, January–March 2025)

Objective: Implement all features iteratively across seven planned development iterations. Each iteration produces tested, integrated software incrementally.

Development Iterations

Iteration	Weeks	Features Implemented
Iteration 1	9–10	SuperTokens OTP auth; Google/Facebook Sign-In; role-based routing; user onboarding
Iteration 2	11	Supabase database schema live; Row Level Security policies; base API structure
Iteration 3	12–13	Therapist directory; therapist profile; admin verification workflow
Iteration 4	13–14	Appointment booking; real-time availability; eSewa payment integration
Iteration 5	15–16	WebRTC video consultation; Socket.IO signalling; admin dashboard
Iteration 6	17–18	Mood tracking; PHQ-9/GAD-7 questionnaires; content hub; Cloudinary integration

Iteration	Weeks	Features Implemented
Iteration 7	18–19	Wellness games; achievement system; subscriptions; push notifications (FCM); offline sync

Artefacts Produced Per Iteration

Artefact Type	Description	Evidence Location
Working software increments	Fully integrated, tested feature increments after each iteration	Git commits; `Project Artefact/04 Development/`
Unit test results	Test results for each unit test as features were completed	`Project Artefact/05 Testing/`
Debugging evidence	Screenshots and logs of bugs found and fixed during development	`Project Artefact/04 Development/`
API endpoint documentation	Updated API reference as new endpoints were added	`07_API_Reference.md`
Code review notes	Self-review notes and refactoring decisions	Weekly artefact logs

Key Technical Decisions Made in Construction

- **GetIt + Injectable:** Dependency injection framework selected to support Clean Architecture layer separation
- **GoRouter:** Navigation with route guards for role-based access control
- **Freezed + json_serializable:** Code generation for immutable state models and JSON parsing
- **Dio:** HTTP client for all API calls with interceptor-based auth token injection
- **flutter_webrtc:** Official WebRTC package for peer connection management
- **Protocol Buffers:** Binary serialisation format for offline-synced mood entries

PHASE 4: TRANSITION (Week 20, March–April 2025)

Objective: Deploy the system, conduct system testing and UAT, gather post-survey data, and complete the final report.

Artefacts Produced

Artefact	Description	Evidence Location
System Test Results (TC-S001–S015)	Results of 15 integration/system test cases	Chapter 4; Appendix K
UAT Plan	Structured UAT plan with 5 target users + 2 external clients	Chapter 4, Section 4.3
UAT Results (TC-UAT001–025)	All 25 UAT test case results including the one partial pass	Chapter 4; Appendix K
Post-Survey (132 responses)	Satisfaction and outcome survey distributed after Transition	Appendix H; `Surveys.md`
User Feedback Forms	UAT feedback forms from all five target user participants	Appendix L
Final Gantt Chart	Actual timeline with deviations noted	Appendix J.1
Client Approval Letters	Signed approval from Ms. Sushana Karki and Dr. Dibyandra Singh	Appendix F
System Screenshots	Captured screenshots of all major screens in deployed state	Appendix J.12
Final FYP Report	Complete academic report submitted for assessment	`FYP Report Writing`
Post-Survey Analysis	100% positive outcomes on all ten post-survey questions	Chapter 4, Section 4.4; Appendix H